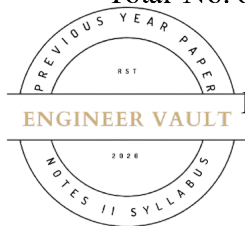


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SUBJECT CODE NO: - RNP-8003**FACULTY OF SCIENCE AND TECHNOLOGY****F.E. (Group A & B) Mech/ Civil / CSE / AI & ML / EEE (NEP) Sem - I&II****Examination June/July 2025****Engineering Physics****[Time: 3 Hours]****[Max. Marks: 60]**

N. B

Please check whether you have got the right question paper.

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q. nos 2, 3, 4, 5, 6 and 7

SECTION A**Q1 Solve the following questions****20**

1. In X-ray emission tubes, X-ray is emitted by the acceleration of
 - a) Atoms
 - b) Protons
 - c) Electrons
 - d) Neutrons
2. The celestial source of X-ray is
 - a) Stars
 - b) Quasars
 - c) Black Holes
 - d) Nebula
3. X-rays have larger wavelengths than which of the following?
 - a) Gamma rays
 - b) Beta rays
 - c) Microwave
 - d) Visible light
4. The critical angle is defined as the angle of incidence at which the angle of refraction is: _____
 - a) 0°
 - b) 90°
 - c) 180°
 - d) 45°
5. Which of the following lenses is used in magnifying glasses?
 - a) Convex lens
 - b) Concave lens
 - c) Cylindrical lens
 - d) Plano-convex lens
6. The free electron theory can explain the following property of metals:
 - a) Low thermal conductivity
 - b) High electrical conductivity
 - c) High elasticity
 - d) Magnetic properties of metals
7. In Newton's Rings, the distance between two consecutive dark rings:
 - a) Increases as we move towards the centre
 - b) Decreases as we move towards the centre
 - c) Remains constant
 - d) Depends on the wavelength of light
8. What is the region enclosed by the optical cavity called?
 - a) Optical Region
 - b) Optical System
 - c) Optical box
 - d) Optical Resonator

- [illegible]

SECTION B

- | | | |
|-----------|---|-----------|
| Q2 | a) Explain the diameter of dark and bright rings. | 05 |
| | b) State and explain piezoelectric effect? | 05 |
| Q3 | a) Explain cyclotron with suitable diagram. | 05 |
| | b) Explain the time dependent Schrodinger equation. | 05 |
| Q4 | a) Explain the ruby Leaser | 05 |
| | b) What is superconductivity? Explain Meissner effect in superconductors? | 05 |
| Q5 | a) Explain Davisson-Germer experiment. | 05 |
| | b) Explain G M counter. | 05 |
| Q6 | a) State and explain spontaneous and stimulated emission. | 05 |
| | b) Write short notes Type -I and Type -II superconductor? | 05 |
| Q7 | a) What is ultrasonic wave and state the properties | 05 |
| | b) Explain He- Ne Leaser with suitable diagram working | 05 |

